

Enterprise Data Contracts on Google Cloud

Shift-Left Governance, Wire Ingress Gates & Dataplex Auto DQ for Financial Systems

Battle-Tested Reference Architecture for Electronic Brokerages & Regulated Exchanges

Cloud Data & Analytics Architecture • Capital Markets Reference Use Case

Governed by Open Data Contract Standard (ODCS v3.0) • Open Reference Implementation

The Core Problem: The "Dump-and-Pray" Pipeline Trap

"In high-frequency and capital markets systems, discovering schema drift downstream is catastrophic."

TRADITIONAL POST-MORTEM MODEL

- **Silent Schema Mutations:** Upstream trading engines alter payloads (e.g. string prices, float volumes), silently corrupting raw data lake tables.
- **Uncontained Blast Radius:** Bad records pollute downstream BigQuery marts, breaking critical surveillance and risk dashboards.
- **Delayed Regulatory Discovery:** Severe SLA violations (e.g. wash trading breaches) are caught weeks late during audit sampling.
- **Costly Remediation:** Engineering teams spend weeks writing bespoke reconciliation scripts, backfilling tables, and re-running pipelines.

SHIFT-LEFT DATA CONTRACT MODEL

- **Perimeter Wire Shield:** Structural schemas are enforced synchronously at the Pub/Sub wire boundary before bad payloads enter the bus.
- **Perimeter Structural Rejection:** Malformed types, missing primary keys, and illegal statuses never touch the streaming bus.
- **Automated DLQ Quarantine:** Storage-level type errors are isolated into Dead-Letter Queues without blocking the highway.
- **In-Database Semantic SLAs:** Continuous Dataplex scans verify financial business rules (anti-wash trading, freshness) in real time.

Open Data Contract Standard (ODCS v3.0): Single Source of Truth

"One declarative contract compiles into wire schemas, analytical DDL, and data quality specifications."

ODCS v3.0 SPECIFICATION PILLARS

- **Identity & Ownership:** Canonical URN (`urn:datacontract:sgx:equity_trades`), contract ownership, and domain scoping.
- **Physical Server Endpoints:** Declarative binding to Pub/Sub topics, schemas, and BigQuery partitioned tables.
- **Strict Data Types:** Field constraints (`required: true` , `primaryKey: true` , logical-to-physical type mapping).
- **Quality Assertions:** Completeness, validity ranges, allowed sets, and custom SQL financial integrity conditions.
- **Service Level Agreements (SLAs):** Strict freshness thresholds (`maxAge: 2h`) anchored to transaction timestamps.

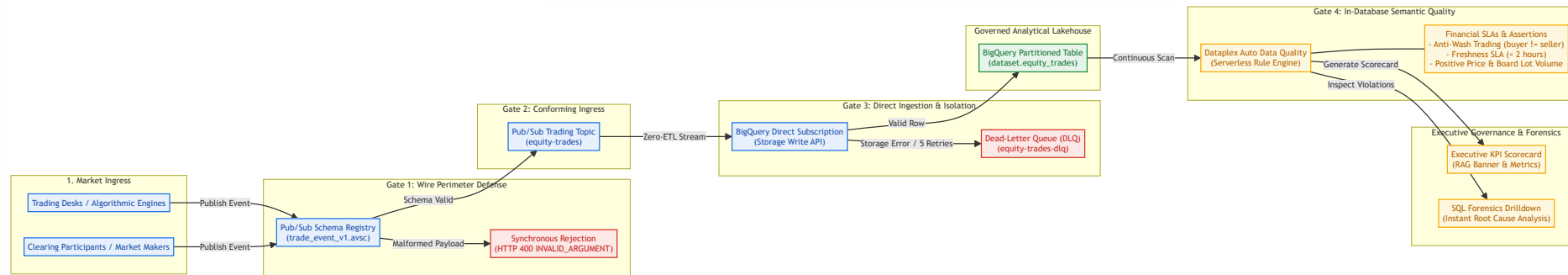
SHIFT-LEFT COMPILER PIPELINE

```
# contract.odcs.yaml (Unified Source of Truth)
apiVersion: 3.0.0
id: urn:datacontract:sgx:equity_trades
servers:
  production_pubsub: { topic: "equity-trades", ... }
  production_bigquery: { dataset: "market_data", ... }
quality:
  - type: sql_assertion
    description: "Anti-wash trading: buyer != seller"
    rowCondition: "buyer_id != seller_id"
```

```
python3 compile_contract.py --contract=contract.odcs.yaml
# Generates:
# 1. schemas/trade_event_v1.avsc      (Pub/Sub Avro)
# 2. sql/create_trades_table.sql      (BigQuery DDL)
# 3. config/dataplex_dq_spec.yaml     (Dataplex DQ)
```

End-to-End Architecture: 4 Defense-in-Depth Enforcement Gates

"Multi-layered defense combining perimeter wire protection with in-database semantic surveillance."



Dual perimeter gates reject wire malformations; BigQuery Storage Write API ingests at millisecond latency; Dataplex executes serverless SQL SLA checks in-database; Executive KPI Scorecard provides continuous governance.

Gates 1 & 2: Wire-Level Perimeter Defense

"Synchronous rejection at the API boundary – stopping invalid payloads before they hit the stream."

PUB/SUB SCHEMA REGISTRY (AVRO JSON)

- **Ingress Wire Gate:** The Pub/Sub topic is bound directly to `trade_event_v1.avsc` using JSON encoding.
- **Synchronous HTTP 400:** Producers attempting to publish non-conforming messages receive immediate `INVALID_ARGUMENT`.
- **Zero Bus Contamination:** Bad events never get assigned a `message_id`, consume zero streaming quota, and generate zero storage cost.
- **Dual-Engine Producer Support:** Fully verified using native `google-cloud-pubsub` client library and transparent `gcloud` subprocess fallback.

DETERMINISTIC TEST VECTORS

Malformed Vector	Injected Flaw	Perimeter Wire Gate Verdict
Missing PK	Omits required <code>trade_id</code>	HTTP 400 INVALID_ARGUMENT
Price Type Mismatch	"price": "MARKET_CLOSE"	HTTP 400 INVALID_ARGUMENT
Volume Type Mismatch	"volume": 100.5	HTTP 400 INVALID_ARGUMENT
Invalid Enum Status	"trade_status": "PENDING"	HTTP 400 INVALID_ARGUMENT

“**Two-Tier Enforcement:** Gate 1 blocks *structural violations* at the wire. Gate 4 catches *semantic violations* (wash trades, price bounds, freshness) in BigQuery.”

Gate 3: BigQuery Direct Ingestion & Dead-Letter Queue (DLQ)

"Zero-ETL streaming ingestion with automated isolation for subtle downstream type failures."

STORAGE WRITE API DIRECT INGESTION

- **Zero-ETL Pipeline:** Native Pub/Sub BigQuery Direct Subscription (`--write-metadata`) streams trades into BigQuery with zero intermediate microservices.
- **Sub-Second Analytics:** Events are available for querying in BigQuery partitioned tables within seconds of execution.
- **Metadata Lineage:** Automatically appends `subscription_name` , `message_id` , `publish_time` , and attributes for compliance audits.
- **Scale & Performance:** Seamlessly handles tens of thousands of trades per second without cluster provisioning.

SUBTLE CORRUPTIONS & DLQ ISOLATION

- **The Type Disparity Trap:** In Avro, `trade_timestamp` is a valid string. In BigQuery DDL, it is `TIMESTAMP NOT NULL` . Passing `"NOT_A_TIMESTAMP"` passes Gate 1 but fails Storage Write API ingestion.
- **Automated Exponential Backoff:** The subscription retries failed writes up to 5 times.
- **Non-Blocking Quarantine:** The failing message is routed to Dead-Letter topic `equity-trades-dlq` and subscriber `equity-trades-dlq-sub` .
- **Stream Continuity:** Main trading event pipeline continues unhindered; corrupted records are preserved for SRE forensics.

Gate 4: In-Database Semantic Quality & Anti-Wash Trading

"Serverless Dataplex Auto Data Quality continuously evaluates deep regulatory business rules."

FINANCIAL INTEGRITY & REGULATORY RULES

- **Anti-Wash Trading Assertion:** Mandated by **FINRA Rule 5210**, **SEC Rule 15c3-5**, and **MAS TRM Section 8**.

```
-- Evaluated inside BigQuery via Dataplex Auto DQ:  
rowCondition: "buyer_id != seller_id"
```

- **Freshness SLA (< 2 Hours):** Validates trade execution records arrive within SLA windows.

```
sqlExpression: "trade_timestamp >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 2 HOUR)"
```

- **Validity Ranges:** Strict enforcement that execution price > 0 and share volume > 0.
- **Completeness Checks:** Zero-null enforcement on `trade_id`, `instrument_code`, `price`, and `volume`.

DATAPLEX AUTO DQ EXECUTION ARCHITECTURE

- **In-Database Compute:** Rules execute directly in BigQuery via serverless pushdown; zero data egress or ETL pipelines.
- **Declarative Spec:** Compiled directly from `contract.odcs.yaml` into `config/dataplex_dq_spec.yaml`.
- **Automated Scheduling:** Can be triggered continuously, on schedule, or via API upon batch arrival.
- **Audit Logging:** Scan results are persisted to Cloud Logging and BigQuery historical governance tables.

Executive Scorecard: Real-Time Governance & Forensics

"Instant visibility for C-Suite risk leaders and deep SQL forensics for data engineering."

EXECUTIVE KPI SCORECARD (RAG BANNER)

```
=====
DATA CONTRACTS: EXECUTIVE DATA QUALITY SCORECARD
=====
>> Executive KPI Dashboard
Target:    ${PROJECT_ID}.sgx_market_data.equity_trades
+-----+-----+-----+-----+
| Metric / Dimension          | Evaluated Value | SLA Threshold | Status |
+-----+-----+-----+-----+
| Overall Table Health Score  | 55.56%         | 100.00%      | FAIL   |
| Completeness SLA Score     | 100.00%        | 100.00%      | PASS   |
| Validity / Range SLA Score  | 33.33%         | 100.00%      | FAIL   |
| Freshness Latency SLA (2h) | 0.00%          | 100.00%      | FAIL   |
| Market Integrity (Anti-Wash)| 0.00%          | 100.00%      | FAIL   |
| Regulatory Compliance (RAG) | RED (CRITICAL: Anti-Wash Breach) | NON-COMPLIANT |
+-----+-----+-----+-----+
```

INSTANT FORENSICS DRILLDOWN QUERIES

Data stewards can immediately run generated debug queries:

- **Anti-Wash Trading Violation Inspection:**

```
SELECT * FROM `sgx_market_data.equity_trades`
WHERE buyer_id = seller_id;
```

- **Freshness SLA Breach Inspection:**

```
SELECT * FROM `sgx_market_data.equity_trades`
WHERE trade_timestamp < TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 2 HOUR);
```

- **Price / Volume Invalidity Inspection:**

```
SELECT * FROM `sgx_market_data.equity_trades`
WHERE price <= 0.0 OR volume <= 0;
```

Turnkey Demonstration: Dual Market Profiles

"Run offline in 60 seconds with zero GCP credentials, or deploy live to Google Cloud."

DUAL MARKET DATA PRESETS

- **global (Default):** Electronic Brokerage profile using liquid US tech equities:
 - Tickers: AAPL , GOOGL , MSFT , NVDA , AMZN
 - Clearing Brokers: BROKER_ALPHA_01 , BROKER_ZETA_02
 - Currency: USD (\$)
- **sgx (Exchange Profile):** Singapore Exchange institutional profile:
 - Tickers: D05.SI , Z74.SI , U11.SI , 039.SI , C6L.SI
 - Clearing Brokers: BROKER_DBS_01 , BROKER_OCBC_02
 - Currency: SGD (S\$)

7-STAGE AUTOMATED EXECUTION

```
# 60-Second Offline Simulation (Global Default)
./run_demo.sh --simulated

# 60-Second Offline Simulation (SGX Profile)
./run_demo.sh --simulated --preset=sgx

# Live Cloud Deployment (when ready to provision)
./setup.sh --project-id=$PROJECT_ID
./run_demo.sh --live --preset=sgx
./cleanup.sh --project-id=$PROJECT_ID
```

“**Comprehensive Test Coverage:** 147 unit, boundary, and adversarial tests passing 100% across all 5 verification tiers.”

Summary & Key Takeaways

SHIFT-LEFT PREVENTATIVE

Enforce at the wire boundary. Blocking malformed structural payloads at Pub/Sub ingress dramatically reduces costly downstream data reconciliation.

DECLARATIVE & NATIVE

1 ODCS contract drives Pub/Sub Avro, BigQuery Storage Write API, and Dataplex Auto DQ. Zero custom ETL pipelines or harvesters to maintain.

AUDIT & REGULATORY READY

Built-in anti-wash trading assertions and freshness SLAs map directly to FINRA Rule 5210, SEC Rule 15c3-5, BCBS 239, and MAS TRM Section 8.

Questions & Discussion

Interactive Demonstration Scripts & Full Source Code: [gcp-data-contracts/](#)

Pre-compiled Slides: [slides/data-contracts-presentation.html](#) & [.pdf](#)